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Profile

Senior ML engineer with 10 years across research and industry, including molecular biology, computational neuroscience, and production ML. Track record owning end-to-end ML work: data engines for long-tailed open-world inference, distributed training infrastructure on Ray, and ML reliability under real-world distribution shift. Currently leading **Ray-bench**, a benchmark and code-RL environment for training coding agents on distributed-systems tasks, at Anyscale.

Machine Learning Experience

Machine Learning Engineer

[Anyscale](#)

San Francisco, CA

Jan. 2025 — Now

Anyscale builds Ray, the open-source distributed compute framework underlying RL post-training, large-scale training, and inference at frontier labs.

- Conceived, pitched, and leading **Ray-bench**: a benchmark for evaluating LLM agents on distributed-systems code (framework translation, Ray code generation from spec, bug fixing, code optimization), paired with **RayGym**, a sandboxed multi-node execution harness for code-RL post-training. Strategic goal: drive Ray adoption among coding agents by giving model providers a verifiable-rewards benchmark to train on. Currently leading data generation: scraping, LLM-assisted synthesis with rejection sampling, AST-based Ray-API verification, and decontamination against pre-training corpora.
- Conceived, pitched, and designing a separate RL environment to train a Ray Core resource auto-calibration model.
- Engineered scalable distributed Ray implementations of: GRPO-based RL post-training for LLMs, LLM-as-judge data curation, 670B-parameter LLM serving, multimodal training, and agentic workflows.
- Early user and feedback partner for Ray LLM and Ray Direct Transport (RDMA-based tensor transfer for disaggregated inference/training); surfaced bugs and informed design.
- Built agentic systems for automated debugging and codebase-wide standards enforcement.
- Designed benchmarks comparing performance and cost across Ray OSS and RayTurbo.

Lead Machine Learning Engineer

[Dendra Systems](#)

Remote

Feb. 2020 — Jan. 2025

Founding ML lead. Built end-to-end ML systems for large-scale, long-tailed visual recognition across thousands of classes — data engine, training infrastructure, evaluation, and serving. Domain: aerial imagery for ecosystem monitoring.

- **Data engine for long-tailed, open-world inference across thousands of target classes.**

- 60% training-data reduction at no performance cost via a novelty-maximizing data-pruning method I developed (original research, unpublished); yielded pareto-optimal (exponential) scaling laws.
 - Active learning to systematically harvest high-leverage training data, preventing model hallucinations on out-of-distribution inputs.
 - Few-shot models paired with an active-learning data-harvesting UI to onboard new classes and domains in hours instead of months (Gradio).
 - Unsupervised “trip-wires” for detecting model hallucinations in production; alerts routed to annotation prioritization workflow (Jira).
 - Annotation QA/QC: systematically surfaced mislabeled and partially labeled samples to create a “self-healing” training dataset.
- **Training and inference infrastructure (Ray-native, cloud-scale).**
 - Distributed, scale-agnostic infrastructure for training, hyperparameter tuning, and inference (Ray/Anyscale, AWS Batch).
 - 20× training-cost reduction via Bayesian hyperparameter tournaments with aggressive early stopping (Ray Tune, HyperOpt, ASHA).
 - Profiled and optimized multi-GPU throughput (Grafana).
 - End-to-end automation of training/inference pipelines (custom orchestrator); later re-implemented serverless for reliability and cost (Step Functions, API Gateway, λ , EventBridge).
 - Researched, experimented, and productionized novel ML techniques: models, samplers, optimization functions, augmentations.
 - **MLOps and reliability.**
 - Sanity checks for detecting silent failures during training; stratified validation across data slices.
 - Debugged model failures using interpretability methods (GradCAM, transformer attention maps) and visualization of training dynamics.
 - Full artifact lineage and reproducibility (MetaFlow).
 - Observability across pipelines (Cloudwatch, Slack bots, UMAP, Sentry); weekly metric reviews to prevent customer-impacting incidents.
 - 80× acceleration in insights delivery via model-in-the-loop annotation tooling.
 - **ML leadership.** Founding ML lead; drove transformation from services to ML-product company. Owned ML roadmap, KPIs, and OKRs aligned with product and operations. ML automation increased throughput and profitability and unlocked new markets. Over-saw data-collection team and ran internal “ML University” lectures.

Lead Data Scientist

[PaceMate](#)

Remote

Jan. 2019 — Dec. 2019

PaceMate monitors transmissions from bluetooth-enabled heart implants, identifying life-threatening arrhythmias and alerting emergency services. Solo ML hire; took cardiac arrhythmia detection from zero to deployed deep-learning system in a regulated, life-critical pipeline — end-to-end ownership across data, modeling, clinician tooling, and SOC2 security.

- Trained SOTA deep neural network for arrhythmia classification, tuned to the implant device used by the majority of patients (Keras).
- Built ECG data-processing pipelines for highly imbalanced clinical data (imbalanced-learn, custom tools).
- Built clinician-facing labeling and prediction-review dashboard for electrophysiologists (Plotly Dash).
- Built EMR analytics dashboard with performant ETL across heterogeneous medical records to enable data-driven decision-making (SQL, PySpark, spaCy, Plotly Dash).
- Translated clinical requirements into ML specs with cardiologists; presented to C-suite and met with investors.

Data Scientist

New College of FL, [F.A.R. Institute](#)

Sarasota, FL

Aug. 2018 — Dec. 2018

Master's capstone (unpaid). Predicted 30-day hospital readmission via patient-visit embeddings (visit2vec, t-SNE); modeled chronic-condition trajectories of heart-failure patients with clustering, network graphs, and finite-state models (SQL, PySpark, NetworkX, TensorFlow).

Research Intern

Peng Lab, [Allen Institute for Brain Science](#)

Seattle, WA

June 2018 — Aug. 2018

Built a 3D Deep Q-Network for tracing neural structures in petabyte-scale fluorescent microscopy; designed the simulation environment and reward system end-to-end based on [manually traced microscope images](#) (TensorFlow, rl-medical, OpenAI Gym; [Profundo writeup](#)).

Research Assistant

Fairhall Lab, [University of Washington](#)

Seattle, WA

Oct. 2014 — Jan. 2016

Computational neuroscience lab. Self-funded via NIH PA-12-149 federal grant.

- Parameter inference: fit biophysical models of mosquito thermonavigation to wind-tunnel behavioral data via numerical optimization (scipy.optimize, sklearn, statsmodels).
- Wrote vectorized numpy routines for high-throughput processing of behavioral video data; computed kinematic and thermal-sensing statistics across the dataset.

Molecular Biologist

Various

Various

Pre-2014

Before transitioning to AI/ML, I was a molecular biology researcher. Co-authored peer-reviewed work on Parkinson's disease (mitochondrial dysfunction in dopaminergic neurons; *PNAS* 2012) and Wnt signaling (*Genetics* 2012). Honors thesis on RNA interference mechanisms in *C. elegans*.

Education

M.S. Data Science

New College of Florida

Sarasota, FL

Aug. 2017 — Dec. 2018

B.A., Chemistry/Biology (with honors)

New College of Florida

Sarasota, FL

Aug. 2007 — May 2011

Early admission (admitted 16 yrs old)
Harriet L. Wilkes Honors College

Jupiter, FL
Jul. 2006 — May 2007

Publications, Presentations, & Teaching

- * 2025 [Deploy DeepSeek-R1 with vLLM and Ray Serve on Kubernetes](#) — end-to-end deployment guide for distributed serving of a 670B-parameter reasoning model.
- * 2025 [Ray Direct Transport: RDMA Support in Ray Core](#) — technical write-up on RDMA-based tensor transfer for disaggregated inference/training.
- * ongoing [Personal blog](#) (80+ technical notes on Ray, LLMs, RL, ML internals, OSS projects).
Selected: [Self-authoring LLM knowledge base](#), [Using DSPy with Ray Data](#), [Slouchless: VLM for posture correction](#).
- * 2025 Speaker at Ray meetups: [Daft+Ray](#), [vLLM+Ray](#), [Skypilot+Ray](#).
- * 2024 Talk at *AI In Production Conference* on [data curation](#).
- * 2021 Invited talk at *Ray Summit*: [How Ray and Anyscale Make it Easy to do Massive-scale ML on Aerial Imagery](#) ([accompanying blog](#)).
- * 2019 Seminar at *New College of FL: Remote Sensing of Cardiac Arrhythmia at Scale using Deep Learning*.
- * 2019 Seminar at *Escuela Secundaria Tecnica de Torquinst: Inteligencia Artificial*.
- * 2015 UW Outreach: various educational events for students from low socioeconomic backgrounds.
- * 2012 [Two peer-reviewed journal articles](#) in *Genetics* and *PNAS*, also presented as posters at three national conferences.
- * 2007 Undergraduate honors thesis on RNA interference mechanisms in *C. elegans*.